Topic: Variation of parameters with the Wronskian

Question: Use variation of parameters and Wronskian integrals to find the general solution to the nonhomogeneous equation.

$$y'' - 5y' + 6y = \frac{1}{e^x}$$

Answer choices:

A $y(x) = c_1e^{2x} + c_2e^{3x} - \frac{5}{12}e^{-x}$

B $y(x) = c_1e^{2x} + c_2e^{3x} + \frac{1}{12}e^{-x}$

C $y(x) = c_1e^{-2x} + c_2e^{3x} - \frac{5}{12}e^{-x}$

D $y(x) = c_1e^{-2x} + c_2e^{3x} + \frac{1}{12}e^{-x}$

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Solution: B

The associated homogeneous equation is

$$y'' - 5y' + 6y = 0$$

so the characteristic will be

$$r^2 - 5r + 6 = 0$$

$$(r - 2)(r - 3) = 0$$

$$r = 2, 3$$

With distinct real roots, the complementary solution is

$$y_c(x) = c_1 e^{2x} + c_2 e^{3x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{2x}, e^{3x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(e^{2x}, e^{3x}) = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix}$$

$$W(e^{2x}, e^{3x}) = (e^{2x})(3e^{3x}) - (e^{3x})(2e^{2x})$$

$$W(e^{2x}, e^{3x}) = 3e^{5x} - 2e^{5x}$$

$$W(e^{2x}, e^{3x}) = e^{5x}$$

Then the particular solution is

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$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -e^{2x} \int \frac{e^{3x}\left(\frac{1}{e^x}\right)}{e^{5x}}\, dx + e^{3x} \int \frac{e^{2x}\left(\frac{1}{e^x}\right)}{e^{5x}}\, dx$$

$$y_p(x) = -e^{2x} \int e^{-3x}\, dx + e^{3x} \int e^{-4x}\, dx$$

$$y_p(x) = -e^{2x}\left(-\frac{1}{3}e^{-3x}\right) + e^{3x}\left(-\frac{1}{4}e^{-4x}\right)$$

$$y_p(x) = \frac{1}{3}e^{-x} - \frac{1}{4}e^{-x}$$

$$y_p(x) = \frac{1}{12}e^{-x}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{12}e^{-x}$$

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Topic: Variation of parameters with the Wronskian

Question: Use variation of parameters and Wronskian integrals to find the general solution to the nonhomogeneous equation.

$$y'' - 4y' + 4y = \frac{e^{2x}}{x^3}$$

Answer choices:

A $y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{e^{2x}}{2x}$

B $y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{e^{2x}}{x^3}$

C $y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{e^{2x}}{x^2}$

D $y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{e^{2x}}{2x^2}$

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Solution: A

The associated homogeneous equation is

$$y'' - 4y' + 4y = 0$$

so the characteristic will be

$$r^2 - 4r + 4 = 0$$

$$(r - 2)(r - 2) = 0$$

$$r = 2, 2$$

With equal real roots, the complementary solution is

$$y_c(x) = c_1 e^{2x} + c_2 x e^{2x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{2x}, xe^{2x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(e^{2x}, xe^{2x}) = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} \end{vmatrix}$$

$$W(e^{2x}, xe^{2x}) = (e^{2x})(e^{2x} + 2xe^{2x}) - (xe^{2x})(2e^{2x})$$

$$W(e^{2x}, xe^{2x}) = e^{4x} + 2xe^{4x} - 2xe^{4x}$$

$$W(e^{2x}, xe^{2x}) = e^{4x}$$

Then the particular solution is

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$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -e^{2x} \int \frac{xe^{2x}\left(\frac{e^{2x}}{x^3}\right)}{e^{4x}}\, dx + xe^{2x} \int \frac{e^{2x}\left(\frac{e^{2x}}{x^3}\right)}{e^{4x}}\, dx$$

$$y_p(x) = -e^{2x} \int \frac{1}{x^2}\, dx + xe^{2x} \int \frac{1}{x^3}\, dx$$

$$y_p(x) = -e^{2x}\left(-\frac{1}{x}\right) - \frac{1}{2}xe^{2x}\left(\frac{1}{x^2}\right)$$

$$y_p(x) = \frac{2e^{2x}}{2x} - \frac{e^{2x}}{2x}$$

$$y_p(x) = \frac{e^{2x}}{2x}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{e^{2x}}{2x}$$

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Topic: Variation of parameters with the Wronskian

Question: Use variation of parameters and Wronskian integrals to find the general solution to the nonhomogeneous equation.

$$y'' - 6y' + 9y = \frac{e^{3x}}{x^5}$$

Answer choices:

A $y(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{e^{3x}}{3x^4}$

B $y(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{e^{3x}}{3x^3}$

C $y(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{e^{3x}}{12x^3}$

D $y(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{e^{3x}}{12x^4}$

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Solution: C

The associated homogeneous equation is

$$y'' - 6y' + 9y = 0$$

so the characteristic will be

$$r^2 - 6r + 9 = 0$$

$$(r - 3)(r - 3) = 0$$

$$r = 3,\ 3$$

With equal real roots, the complementary solution is

$$y_c(x) = c_1 e^{3x} + c_2 x e^{3x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{3x}, xe^{3x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(e^{3x}, xe^{3x}) = \begin{vmatrix} e^{3x} & xe^{3x} \\ 3e^{3x} & e^{3x} + 3xe^{3x} \end{vmatrix}$$

$$W(e^{3x}, xe^{3x}) = (e^{3x})(e^{3x} + 3xe^{3x}) - (xe^{3x})(3e^{3x})$$

$$W(e^{3x}, xe^{3x}) = e^{6x} + 3xe^{6x} - 3xe^{6x}$$

$$W(e^{3x}, xe^{3x}) = e^{6x}$$

Then the particular solution is

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$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -e^{3x} \int \frac{xe^{3x}\left(\frac{e^{3x}}{x^5}\right)}{e^{6x}}\, dx + xe^{3x} \int \frac{e^{3x}\left(\frac{e^{3x}}{x^5}\right)}{e^{6x}}\, dx$$

$$y_p(x) = -e^{3x} \int \frac{1}{x^4}\, dx + xe^{3x} \int \frac{1}{x^5}\, dx$$

$$y_p(x) = -e^{3x}\left(-\frac{1}{3}x^{-3}\right) + xe^{3x}\left(-\frac{1}{4}x^{-4}\right)$$

$$y_p(x) = \frac{e^{3x}}{3x^3} - \frac{e^{3x}}{4x^3}$$

$$y_p(x) = \frac{e^{3x}}{12x^3}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{e^{3x}}{12x^3}$$

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